

Leona Francis

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OBJECTIVE

Highly motivated Computer Science undergraduate with hands-on institutional research experience in Deep Learning and Computer Vision. Seeking a **Student Researcher** role at Google to contribute to exploratory research projects, advance intelligent systems, and develop scalable solutions for real-world, large-scale problems.

EDUCATION

B.S. IN COMPUTER SCIENCE | MORGAN STATE UNIVERSITY | GRADUATION: 05/29

PROJECTS

DermaBridge: Multimodal Clinical Screening Application

- **Engineered a synchronized data pipeline using Python** to clean, normalize, and augment unstructured dermatoscopic scans and structured patient metadata across 3 major medical repositories.
- **Integrated state-of-the-art computer vision models**, benchmarking **Swin Transformers**, **EfficientNetV2**, and **traditional CNNs** to optimize hierarchical feature extraction and surpass legacy baseline classification accuracy.
- **Deployed algorithmic bias mitigation frameworks** into the training cycle, utilizing skin tone distribution analysis to ensure equitable diagnostic performance across diverse demographic groups.
- **Implemented XAI (Explainable AI) tools**, embedding **Grad-CAM** and **SHAP** to generate visual heatmaps and feature-based justifications for transparent, clinician-ready model predictions.
- Built a functional full-stack prototype application that handles end-to-end patient screening, confidence-aware risk profiling, and location-based medical referral routing.

Kode with Klossy workshop

- **Managed the full software development lifecycle** in a 4-member team, from initial concept to deployment.
- Developed and implemented key modules, including user authentication (Sign Up/Login) and curated web-view navigation, while applying modern UI/UX design principles.
- Engineered a functional mobile application during an intensive two-week workshop, demonstrating commitment and rapid delivery.

Python Scripting & Data Processing

- **Designed an automated Python process** to perform iterative data processing and bulk calculations (e.g., summing, averaging a defined range of numbers).
- Implemented conditional logic to prevent critical errors (e.g., division-by-zero errors), demonstrating a detail-oriented approach to data quality checks.

EXPERIENCE

Student Researcher | CEAMLS Sairi Program | Morgan State University | May 2026 – July 2026

- **Developed and benchmarked multimodal deep learning models** combining unstructured dermatoscopic images and patient metadata from global datasets (HAM10000, PAD-UFES-20, MRA-MIDAS) to optimize skin lesion classification.

- **Upgraded system architecture** by implementing and comparing **EfficientNetV2** and Vision Transformers against prior baseline models, significantly improving classification robustness and deployment readiness.
- **Prioritized health equity** by analyzing skin tone distribution across datasets and integrating a comprehensive fairness framework to minimize demographic performance gaps and algorithmic bias.
- **Embedded an explainability pipeline** using **Grad-CAM** and **SHAP** alongside confidence-aware logic to provide clinicians with transparent, interpretable risk feedback.
- **Designed and prototyped a functional mobile/web application** that delivers end-to-end skin lesion screening, automated risk assessments, and location-aware clinical referral support.

Peer Career Ambassador / Morgan State University / October 2025 – Current

- **Mentored an average of 10 students per week**, providing targeted, data-informed feedback to significantly improve career materials.
- **Developed and executed on-campus promotional strategies** (presentations, social media), requiring the ability to translate information into clear, actionable messages for broad engagement.
- **Managed logistics** for large-scale career fairs, demonstrating strong collaboration and process management skills.

Concept Attendant / PASCHAL'S / Atlanta, GA / April 2025 – August 2025

- Facilitated team efficiency in a high-volume environment by adapting to multiple roles.
- Enhanced 15+ guest experience through clear communication and conscientious service.

SKILLS & ABILITIES

- **Programming & Data:** Python, Swift.
- **Machine Learning & AI:** Multimodal Deep Learning, Computer Vision, CNNs, Vision Transformers (ViT), EfficientNetV2
- **Data Engineering:** Data Curation, Image Normalization, Augmentation, Metadata Cleaning, Fairness/Bias Mitigation Frameworks, Data Quality Checks

INVOLVEMENT & LEADERSHIP

- CyberX – Cybersecurity Organization (2025 – Present): Participating in ethical hacking and network defense workshops.
- AI Club Engineers (2025 – Present): Actively learning AI and machine learning applications, providing foundational knowledge for prototyping ML workflows.